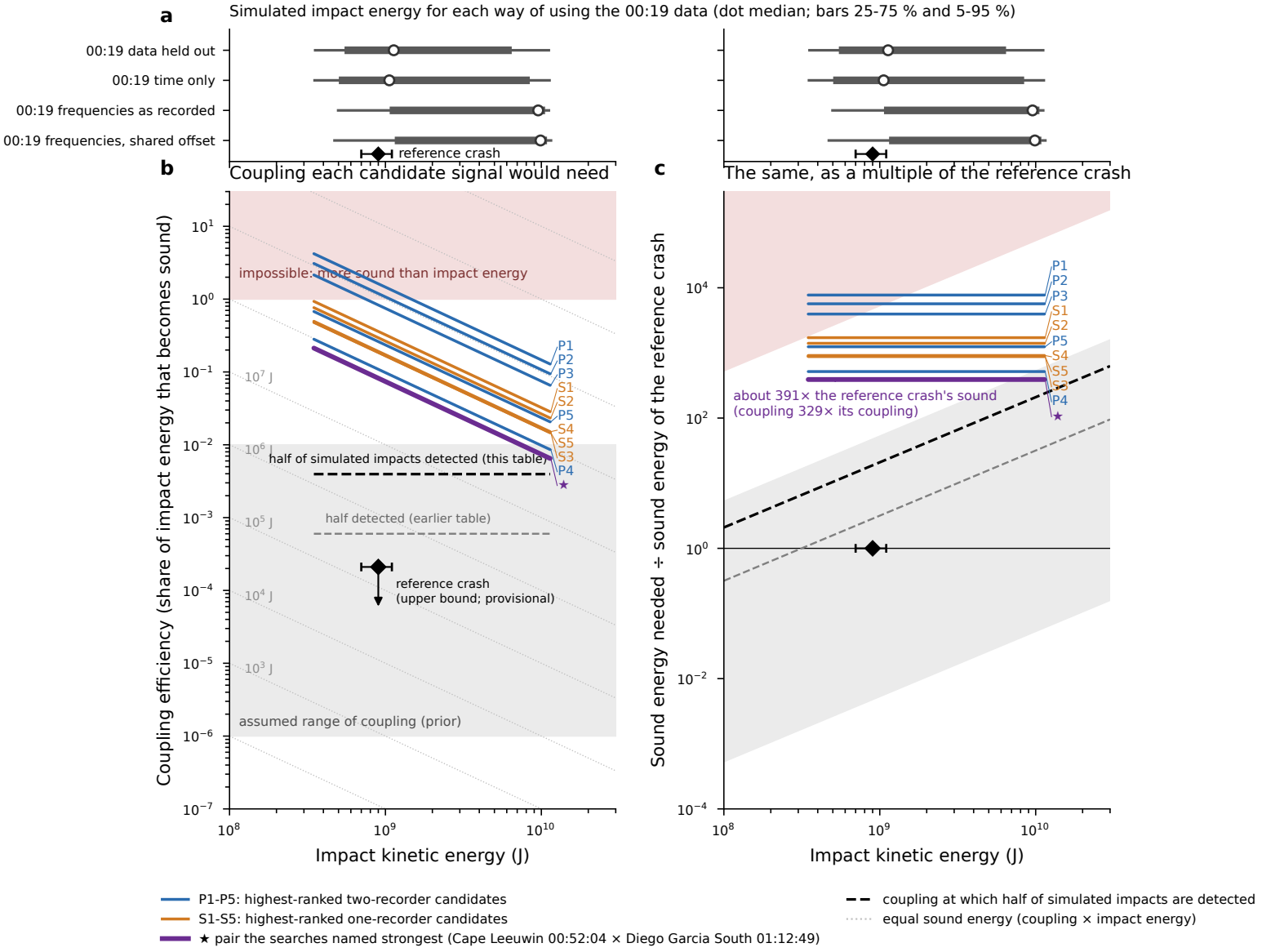
The candidate signals need far more of the impact energy to become sound than the reference crash did



(i) Each line is one candidate signal. It shows the share of the impact energy that must become sound to make that signal, for impact energies from the 5 % to the 95 % value of the simulation. Above 1 is not possible. The reference crash (a military jet that hit the sea) changed only a small share of its energy into sound; that value is provisional. The ★ pair needs about 330 times the coupling of the reference crash, at the simulated impact energy of about 1.1 GJ; because that energy is about 1.2 times the crash energy (0.9 GJ), the sound energy needed is about 390 times the crash's sound energy. Its sound direction does not agree with the directions Kadri published. These are estimates, not detections.

(ii) Source posterior: run J geometry sample (8.85 M rows), 00:19 Held Out (D0'), p = 0.5; energy = contact KE (energy_transferred_j = 1/2 m v^2). Bands: eta(E) = R/E with R = eta_needed x E_ref (pairs: eta for P_D(both) = 0.5, alpha 0.05, at the lag's arc locus with run J median E within +/-1 deg; singles: level match at the reference impact 37.1 S, 1.13 GJ, tau90 3.3 s). F-35A: E = 900 +/- 200 MJ, eta = 2.1e-4 (provisional - from Brown 2026's quote of Metz 2023; re-derivation from Metz pending); an upper bound (shallow sound channel). Half-detected line: P_D(any of 6 receivers) = 0.5 at alpha 0.01 with the +/-10 dB calibration draw, recomputed with the block 11 machinery on the corrected IMOS RAM-grid table: eta50 = 4.0e-03 (alpha 0.05: 1.1e-03); earlier table 6e-4 (injection-recovery note). Star pair: H01W 00:52:04 M1b p 0.043 (digitised Kadri Fig. 9c; text time 00:52:03, caption time 00:54:30) x H08S 01:12:49 M1b p 0.039 (raw panel e; p_B 0.30, airgun-dominated), lag 1,245 s in [809, 1536]; locus 33.6 S 94.5 E, impact 00:31:41; locus bearing 268.9 deg from H01W, 144.8 deg from H08S; Kadri gives 57 or 306.18 deg (text p. 9 vs Table 1 caption), neither within 37 deg; nearest Table 1 bearing 00:52:50 at 274.84 deg (46 s later). Lines (ranked by approximate ln LR, ranked_pairs.csv / ranked_singles.csv): P1 H08S 01:02:53 x 3315 00:47:52; P2 H08S 01:03:13 x 3315 00:47:52; P3 H08S 01:02:53 x 3315 00:47:16; P4 H08S 01:04:52 x 3315 00:49:48; P5 H08S 01:04:23 x 3315 00:49:48; S1 H08S 01:02:53 M1b; S2 3275 01:20:35 M1b; S3 3275 01:07:46 D1; S4 3315 00:47:16 BE; S5 3275 01:07:47 M1b. tau dependence of the required level ignored along each band. EXPLORATORY; PREDICTIVE, NOT CALIBRATED; caveats C1 (+10.4/+7.4 dB per octave slope misfit at H01W/H08S), C2.